Advanced Excel Lab Assisgnment 3

Student Name	Math	Science	English	History	Total Score	Average Score
Pari Agarwal	85	78	92	88	?	?
Rakshit Sharma	90	82	88	85	?	?
Krishn Mishra	78	75	85	80	?	?
Vanshika Gupta	92	88	90	94	?	?
Saurabh Dahiya	75	70	78	72	?	?

Average score in Maths	?
Average score in Science	?
Average score in History	?
Average score in English	?

Average score across all subjects	?

Number of Subject	?
Number of Student	?
Maximum Marks in All Subejcts	?
Minimum Marks in All Subejcts	?

Advanced Excel Lab Assisgnment 4

Q.No	Question
1	Sort the Sales column in descending order (Z→A).
2	Find the product with the highest sales.
3	Find the product with the lowest sales.
4	Apply a filter to show only products with sales greater than 500.
5	Count how many products have sales greater than 500.
6	Display only products with sales less than 1000.
7	Find the top 5 products by sales.
8	Find the bottom 5 products by sales.
9	Apply a filter to show sales between 700 and 1200.
10	Find the average sales of all products.
11	Find the total sales of all products.
12	Find how many products have sales greater than 1500.
13	Highlight products with sales more than 1000.
14	Sort the products alphabetically (A→Z).
15	Find the 10th highest sales value.
16	Find the 5th lowest sales value.
17	Apply filter to display only even-numbered sales values.
18	Apply filter to display only odd-numbered sales values.
19	Find the median sales value.
20	Filter products with sales equal to exactly 1000.

Product and Sales Data

Product	Sales
Product_1	1239
Product_2	115
Product_3	1814
Product_4	344
Product_5	903
Product_6	535
Product_7	776
Product_8	400
Product_9	1477
Product_10	1138
Product_11	1644
Product_12	1730
Product_13	636
Product_14	1544
Product_15	1502
Product_16	812
Product_17	1306
Product_18	1524
Product_19	874
Product_20	1192
Product_21	170
Product_22	670
Product_23	1901
Product_24	1127
Product_25	1589
Product_26	1825
Product_27	1396
Product_28	723
Product_29	1036
Product_30	335

Q.No	Question	Solution
1	Sort the Sales column in descending order (Z→A).	Go to Data → Sort → Sales → Largest to Smallest
2	Find the product with the highest sales.	The first product after sorting Z→A
3	Find the product with the lowest sales.	The last product after sorting Z→A
4	Apply a filter to show only products with sales greater than 500.	Data → Filter → Number Filters → Greater than 500
5	Count how many products have sales greater than 500.	Check the filtered result count at the bottom left of Excel
6	Display only products with sales less than 1000.	Filter → Number Filters → Less than 1000
7	Find the top 5 products by sales.	After sorting Z→A, take the first 5 rows
8	Find the bottom 5 products by sales.	After sorting A→Z, take the first 5 rows
9	Apply a filter to show sales between 700 and 1200.	Number Filters → Between → 700 and 1200
10	Find the average sales of all products.	Use =AVERAGE(B2:B31)
11	Find the total sales of all products.	Use =SUM(B2:B31)
12	Find how many products have sales greater than 1500.	Use =COUNTIF(B2:B31,">1500")
13	Highlight products with sales more than 1000.	Conditional Formatting → Highlight Cell Rules → Greater Than 1000
14	Sort the products alphabetically (A→Z).	Data → Sort → Product → A to Z
15	Find the 10th highest sales value.	Use =LARGE(B2:B31,10)
16	Find the 5th lowest sales value.	Use =SMALL(B2:B31,5)
17	Apply filter to display only even-numbered sales values.	Add helper column with =ISEVEN(B2) → Filter TRUE
18	Apply filter to display only odd-numbered sales values.	Add helper column with =ISODD(B2) → Filter TRUE
19	Find the median sales value.	Use =MEDIAN(B2:B31)
20	Filter products with sales equal to exactly 1000.	Filter → Number Filters → Equals → 1000

Advanced Excel Lab Assisgnment 5

Q No.	Question
1	Use CONCATENATE to join First Name and Last Name into Full Name.
2	Use TEXTJOIN with a space delimiter to combine First Name and Last Name.
3	Extract the first 3 letters of the First Name using LEFT.
4	Extract the last 2 letters of the Last Name using RIGHT.
5	Extract 4 characters starting from the 2nd character of the Last Name using MID.
6	Convert the First Name into UPPER case.
7	Convert the Last Name into lower case.
8	Convert the Department name into Proper case.
9	Count total characters in the City name using LEN.
10	Remove extra spaces from First Name using TRIM.
11	Create an email ID in format firstname.lastname@company.com using CONCATENATE.
12	Use LEFT to extract first letter of First Name (for initials).
13	Use RIGHT to extract last letter of Last Name.
14	Combine Initials with Last Name using CONCATENATE.
15	Use MID to extract middle part of Department name.
16	Use PROPER to capitalize names correctly.
17	Use UPPER for department names.
18	Use LOWER for city names.
19	Join First Name, Last Name, and Department using TEXTJOIN with commas.
20	Count length of Full Name using LEN.
21	Create Username as firstname+first letter of last name using CONCATENATE.
22	Trim spaces and then concatenate names.
23	Use TEXTJOIN to create mailing labels with City and Department.
24	Extract first 2 letters of City using LEFT.
25	Extract last 3 letters of Department using RIGHT.
26	Use MID to extract letters 2-5 of First Name.
27	Convert Email ID into lowercase using LOWER.
28	Proper case City names using PROPER.
29	Count characters of Email IDs using LEN.
30	Remove spaces from Department and then combine with City.

Advanced Excel Lab Assisgnment 6

Student Name	Marks	Result	Grades
Praveen Kumar Sharma	55	?	?
Akshat Agrawal	38	?	?
NAMAN NARWANI	45	?	?
shobhit verma	35	?	?
Aryan Singh Sikarwar	60	?	?
HIMANSHU PALIWAL	42	?	?
Vaibhav Saini	87	?	?
Lalit Kumar	67	?	?
Yash Kharbanda	77	?	?
Tanush Kumawat	57	?	?
Sanjeev Shrivastava	47	?	?

Question: Write PASS if marks greather than 50 else Fail

Label Grades Based on Marks:

Extend the grading system to include grades:

80 and above = “A”

60 to 79 = “B”

40 to 59 = “C”

Below 40 = “Fail”

Advanced Excel Lab Assisgnment 7

Task 1: Calculate the number of complete YEARS of service for each employee.
Formula Example: =DATEDIF(C2,D2,"Y")

Task 2: Calculate the number of remaining MONTHS (ignoring years).
Formula Example: =DATEDIF(C2,D2,"YM")

Task 3: Calculate the number of remaining DAYS (ignoring months and years).
Formula Example: =DATEDIF(C2,D2,"MD")

Task 4: Create a column that shows FULL SERVICE PERIOD in the format:
Example: '10 Years, 3 Months, 12 Days'
Hint: Use CONCATENATE or TEXTJOIN with DATEDIF results.

Task 5: Identify employees who have completed more than 10 years of service.
Hint: Use IF condition with DATEDIF in Years column.

Task 6: Calculate how many employees joined before 2015.
Hint: Use COUNTIF with Joining_Date.

Task 7: Create a summary table showing:
- Number of employees with < 5 years service
- Number of employees with 5-10 years service
- Number of employees with > 10 years service

Employee_ID	Employee_Name	Joining_Date	Current_Date
E101	Rahul Sharma	2018-12-03 00:00:00	2025-09-17
E102	Priya Mehta	2018-03-30 00:00:00	2025-09-17
E103	Amit Verma	2011-10-16 00:00:00	2025-09-17
E104	Sneha Kapoor	2018-11-03 00:00:00	2025-09-17
E105	Vikas Singh	2017-10-28 00:00:00	2025-09-17
E106	Anjali Nair	2017-12-27 00:00:00	2025-09-17
E107	Rohit Gupta	2013-11-02 00:00:00	2025-09-17
E108	Kiran Rao	2019-03-09 00:00:00	2025-09-17
E109	Manish Yadav	2020-05-24 00:00:00	2025-09-17
E110	Neha Bhatia	2021-01-28 00:00:00	2025-09-17
E111	Sandeep Joshi	2022-09-04 00:00:00	2025-09-17
E112	Pooja Jain	2012-01-04 00:00:00	2025-09-17
E113	Arjun Malhotra	2021-08-26 00:00:00	2025-09-17
E114	Divya Chauhan	2012-12-19 00:00:00	2025-09-17
E115	Kunal Mishra	2016-10-14 00:00:00	2025-09-17
E116	Simran Kaur	2021-05-06 00:00:00	2025-09-17
E117	Harsh Vardhan	2017-03-19 00:00:00	2025-09-17
E118	Nikita Roy	2014-09-19 00:00:00	2025-09-17
E119	Aditya Khanna	2013-08-19 00:00:00	2025-09-17
E120	Meera Iyer	2016-11-25 00:00:00	2025-09-17

Advanced Excel Lab Assisgnment 8

Conditional Formatting Questions

1. Highlight all employees whose **Salary is greater than 9500** in light green.

2. Apply a **red fill** to employees in the **Admin department** with a salary less than 9500.

3. Use **data bars** to visually represent salaries in the **Salary** column.

4. Highlight employees from the **'east' region** in yellow.

5. Apply a **3-color scale** to the Salary column, with lowest in red, middle in yellow, and highest in green.

6. Highlight employees whose **First Name starts with 'S'**.

7. Highlight rows where **Hiredate is before 01-Jan-1990**.

8. Highlight employees in the **CCD department** with Salary exactly **9000**.

9. Highlight the **top 3 highest salaries** in blue.

10. Highlight the **bottom 3 salaries** in orange.

11. Shade all employees who have **Deptcode 70** in pink.

12. Use **icon sets** (traffic lights) to represent Salary values.

13. Highlight employees whose **First Name and Last Name length combined is greater than 12 characters**.

14. Highlight male employees earning **more than 9500** in green and female employees earning **less than 9500** in red.

15. Highlight employees whose **Hiredate year is 1988**.

16. Highlight employees whose **Last Name contains 'a'** (case-insensitive).

17. Highlight alternate rows in light gray for better readability.

18. Highlight employees from **Admin** department who were hired **after 1990**.

19. Highlight employees whose salary is **above the average salary** in light blue.

20. Highlight employees in **Region = east** and **Salary < 9500** in red.

HR Data

Empcode	First Name	Last Name	Gender	Dept	Region	Deptcode	Hiredate	Salary
74	Suraj	Saksena	Male	Admin	east	70	25-Oct-88	#######
16	Aakash	Dixit	Female	Admin	east	70	1-Mar-83	9,000.00
64	Timsi	Desai	Female	CCD	east	50	26-Oct-88	#######
53	Vishal	Virsinghani	Male	CCD	east	50	3-Mar-95	9,000.00
54	Lalita	Rao	Female	CCD	east	50	1-Nov-86	9,000.00
12	Sheetal	Desai	Female	Director	east	80	12-Dec-84	#######
9	Sujay	Madhrani	Male	Finance	east	40	21-Dec-85	8,500.00
92	Payal	Singhani	Male	Mktg	east	20	5-Apr-89	9,900.00
69	Reeta	Naik	Female	Mktg	east	20	4-Mar-95	8,000.00
3	Beena	Mavadia	Female	Mktg	east	20	24-Nov-79	7,000.00
20	Suchita	Panchal	Male	Mktg	east	20	7-Jun-99	4,500.00
41	Piyush	Surti	Female	Personal	east	60	19-Oct-88	9,000.00
5	Deepak	Jain	Female	Personal	east	60	17-Aug-90	7,900.00
81	Kalpana	Shirishkar	Male	R&D	east	30	1-Nov-88	#######
29	Giriraj	Gupta	Male	R&D	east	30	1-Oct-82	9,000.00
40	Pooja	Gokhale	Female	R&D	east	10	29-Sep-91	7,600.00
84	Pinky	Robert	Female	R&D	east	30	3-Mar-99	4,000.00
91	Niki	Digaria	Male	Sales	east	10	13-Nov-79	#######
73	Nita	Pandhya	Female	Sales	east	10	24-Oct-88	#######
25	Rishi	Malik	Female	Sales	east	10	2-Jul-92	7,500.00
60	Anuradha	Zha	Male	Admin	north	70	25-Nov-87	#######
15	Aalok	Trivedi	Male	Admin	north	70	1-Mar-83	9,000.00
46	Vicky	Joshi	Female	Admin	north	70	7-Feb-88	9,000.00
2	Kuldeep	Sharma	Female	Admin	north	70	1-Mar-99	4,000.00
51	Heena	Godbole	Female	CCD	north	50	21-Oct-88	#######
72	Laveena	Shenoy	Female	CCD	north	50	23-Oct-88	#######
86	Sonia	Sasan	Female	CCD	north	50	15-Jan-98	#######
98	Chetan	Dalvi	Male	CCD	north	50	11-Apr-89	9,900.00
30	Ankur	Joshi	Male	CCD	north	50	1-Oct-82	9,000.00
32	Zarina	Vora	Female	CCD	north	50	29-Sep-91	5,000.00
83	Kunal	Shah	Female	CCD	north	50	2-Mar-99	4,000.00
37	Sheetal	Dodhia	Male	Finance	north	40	1-Oct-82	#######
11	Meera	Lalwani	Male	Finance	north	40	11-Dec-84	#######
57	Maya	Panchal	Female	Mktg	north	20	5-Nov-90	#######
78	Sagar	Bidkar	Female	Mktg	north	20	29-Oct-88	#######
79	Dayanand	Gandhi	Male	Mktg	north	20	30-Oct-88	#######
8	Andre	Fernendes	Female	Mktg	north	20	20-Jul-77	9,000.00
33	Arun	Joshi	Female	Mktg	north	20	5-May-92	5,300.00
19	Satinder Kaur	Sasan	Male	Mktg	north	20	6-Jun-99	4,500.00
18	Farhan	Sadiq	Male	Mktg	north	20	5-Jun-99	3,400.00
66	Uday	Naik	Female	Personal	north	60	28-Oct-88	#######
65	Parul	Shah	Female	Personal	north	60	27-Oct-88	#######
14	Priya	Shirodkar	Male	Personal	north	60	14-Dec-84	8,500.00
28	Aalam	Qureshi	Female	Personal	north	60	19-Oct-88	8,100.00
68	Pravin	Joshi	Female	Personal	north	60	3-Mar-95	8,000.00
90	Priyanka	Mehta	Male	R&D	north	30	6-Jan-80	#######
4	Seema	Ranganathan	Male	R&D	north	30	4-Sep-89	#######
62	Waheda	Sheikh	Female	R&D	north	30	24-Oct-88	#######
76	Drishti	Shah	Female	R&D	north	30	27-Oct-88	#######
6	Neena	Mukherjee	Male	R&D	north	30	4-Sep-89	7,100.00
44	Kinnari	Mehta	Female	R&D	north	30	7-Jul-97	6,500.00
85	Ruby	Joseph	Female	R&D	north	30	14-Jan-98	4,000.00
89	Tejal	Patel	Male	Sales	north	10	21-Feb-79	#######
1	Raja	Raymondekar	Female	Sales	north	10	1-Jan-77	#######

HR Data

Empcode	First Name	Last Name	Gender	Dept	Region	Deptcode	Hiredate	Salary
61	Asha	Trivedi	Female	Sales	north	10	26-Nov-87	#######
45	Jeena	Baig	Male	Sales	north	10	29-Sep-91	9,000.00
50	Mario	Fernandes	Female	Sales	north	10	20-Oct-88	9,000.00
7	Pankaj	Sutradhar	Female	Sales	north	10	12-Dec-99	8,500.00
67	Mandakini	Desai	Male	Sales	north	10	1-Dec-95	8,000.00
38	Richa	Raje	Male	Sales	north	10	29-Sep-91	6,000.00
58	Disha	Parmar	Female	Admin	south	70	14-Jan-87	#######
39	Kirtikar	Sardesai	Female	Admin	south	70	14-Jun-97	4,500.00
80	Tara	Phule	Male	CCD	south	50	31-Oct-88	#######
48	Rakesh	Kumar	Female	CCD	south	50	19-Oct-88	9,000.00
31	Tapan	Ghoshal	Female	CCD	south	50	7-Jul-97	4,000.00
36	Chitra	Pednekar	Male	Finance	south	40	1-Oct-82	#######
100	Rupesh	Sawant	Male	Finance	south	40	2-Feb-99	4,500.00
59	Geeta	Darekar	Female	Mktg	south	20	24-Nov-87	#######
87	Jignesh	Tripathi	Female	Mktg	south	20	16-Jan-98	#######
95	Kabir	Vora	Male	Mktg	south	20	8-Apr-89	9,900.00
63	Veena	Patil	Male	Personal	south	60	25-Oct-88	#######
13	K. Sita	Narayanan	Female	Personal	south	60	13-Dec-84	8,500.00
34	Pooja	Khetan	Female	Personal	south	10	3-Sep-94	6,700.00
71	Yamini	Gupta	Male	R&D	south	30	22-Oct-88	#######
97	Pushpa	Raut	Female	R&D	south	30	10-Apr-89	9,900.00
26	Mala	Bhaduri	Female	R&D	south	30	19-Mar-95	6,000.00
24	Bharat	Shetty	Male	Sales	south	10	1-Oct-82	#######
2	Suman	Shinde	Male	Sales	south	10	1-Jan-77	#######
93	Harsha	Trivedi	Female	Sales	south	10	6-Apr-89	9,900.00
56	Katti	Surti	Female	Sales	south	10	9-May-93	8,500.00
10	Shilpa	Lele	Male	Admin	west	70	1-Mar-83	#######
27	Hajra	Hoonjan	Female	Admin	west	70	4-May-96	5,500.00
75	Nayeem	Khan	Female	CCD	west	50	26-Oct-88	#######
47	Neha	Joshi	Male	CCD	west	50	19-Oct-88	9,000.00
52	Mehul	Sheth	Male	CCD	west	50	1-Dec-95	9,000.00
35	Shilpa	Parikh	Female	Finance	west	40	1-Oct-82	#######
99	Indu	Shah	Female	Finance	west	40	30-Dec-97	6,000.00
94	Radhika	Kulkarni	Male	Mktg	west	20	7-Apr-89	9,900.00
17	Parvati	Khanna	Female	Mktg	west	20	13-Aug-86	6,000.00
21	Shazia	Sheikh	Female	Mktg	west	20	8-Jun-99	4,500.00
88	Vinit	Shrivastava	Male	Mktg	west	20	30-Dec-97	4,500.00
49	Ruheal	Baig	Male	Personal	west	60	19-Oct-88	9,000.00
42	Shaheen	Khan	Female	Personal	west	60	17-Nov-90	7,900.00
96	Beena	Sharma	Female	R&D	west	30	9-Apr-89	9,900.00
70	Surendra	Godse	Male	R&D	west	30	5-Mar-95	8,000.00
5	Julie	D'Souza	Female	R&D	west	30	4-Sep-88	7,100.00
23	Jasbinder	Khurana	Male	R&D	west	30	24-Apr-99	4,500.00
77	neha	Chhaya	Female	Sales	west	10	28-Oct-88	#######
22	Pooja	Nimkar	Female	Sales	west	10	########	8,500.00
55	Kajal	Joglekar	Female	Sales	west	10	1-Oct-92	8,500.00
101	Rachana	Chotaliya	Female	training	west	90	########	#######
70	Surendra	Godse	Male	R&D	west	30	5-Mar-95	#######
49	Ruheal	Baig	Male	Personal	west	60	19-Oct-88	9,000.00
42	Shaheen	Khan	Female	Personal	west	60	17-Nov-90	7,900.00
96	Beena	Sharma	Female	R&D	west	30	9-Apr-89	9,900.00
70	Surendra	Godse	Male	R&D	west	30	5-Mar-95	8,000.00

Step-by-step solutions

1) Salary > 9500 (light green)

Range: `A2:I{N}`
Home → Conditional Formatting → New Rule → Use a formula to determine which cells to format
Formula: `=$I2>9500`
Format: Fill = light green → OK.

2) Admin dept with Salary < 9500 (red)

Range: `A2:I{N}`
New Rule → Use a formula
Formula: `=AND($E2="Admin",$I2<9500)`
Format: Fill = red → OK.

3) Salary Data Bars

Range: `I2:I{N}`
Home → Conditional Formatting → Data Bars → (choose any Gradient or Solid Fill).

4) Region = east (yellow)

Range: `A2:I{N}`
New Rule → Use a formula
Formula: `=LOWER($F2)="east"`
Format: Fill = yellow → OK.

5) 3-Color Scale on Salary

Range: `I2:I{N}`
Home → Conditional Formatting → Color Scales → 3-Color Scale (Green-Yellow-Red).
(You can customize min/mid/max under "Manage Rules" if needed.)

6) First Name starts with "S"

Range: `A2:I{N}`
New Rule → Use a formula
Formula: `=LEFT(UPPER($B2),1)="S"`
Format: Choose a highlight (e.g., light blue) → OK.

7) Hiredate before 01-Jan-1990

Range: `A2:I{N}`
New Rule → Use a formula
Formula: `=$H2<DATE(1990,1,1)`
Format: Choose a highlight (e.g., light orange) → OK.

8) CCD dept with Salary exactly 9000

Range: `A2:I{N}`
New Rule → Use a formula
Formula: `=AND($E2="CCD",$I2=9000)`
Format: Choose a distinct fill (e.g., teal) → OK.

9) Top 3 highest salaries (blue)

Range: `I2:I{N}`
Home → Conditional Formatting → Top/Bottom Rules → Top 10 Items… → change 10 to 3
Format: Fill = blue (or any) → OK.

10) Bottom 3 salaries (orange)

Range: `I2:I{N}`
Home → Conditional Formatting → Top/Bottom Rules → Bottom 10 Items… → change 10 to 3
Format: Fill = orange → OK.

11) Deptcode = 70 (pink)

Range: `A2:I{N}`
New Rule → Use a formula
Formula: `=$G2=70`
Format: Fill = pink → OK.

12) Icon Sets for Salary (traffic lights)

Range: `I2:I{N}`
Home → Conditional Formatting → Icon Sets → choose a 3-icon set (e.g., 3 Traffic Lights).
(Optional: Manage Rules → Edit Rule to set "Show Icon Only" or adjust thresholds.)

13) First + Last Name length > 12 characters

Range: `A2:I{N}`
New Rule → Use a formula
Formula: `=LEN($B2&$C2)>12`
Format: Choose a highlight → OK.

14) Two rules:

(a) Male with Salary > 9500 (green)

- **Range:** `A2:I{N}`
- **New Rule → Use a formula**
- **Formula:** `=AND($D2="Male",$I2>9500)`
- **Format:** Fill = green → OK.

(b) Female with Salary < 9500 (red)

- **Range:** `A2:I{N}`
- **New Rule → Use a formula**
- **Formula:** `=AND($D2="Female",$I2<9500)`
- **Format:** Fill = red → OK.

(Ensure the rule order doesn't conflict; they're independent.)

15) Hire year is 1988

Range: `A2:I{N}`
New Rule → Use a formula
Formula: `=YEAR($H2)=1988`
Format: Choose a highlight → OK.

16) Last Name contains "a" (case-insensitive)

Range: `A2:I{N}`
New Rule → Use a formula
Formula: `=ISNUMBER(SEARCH("a",$C2))`
Format: Choose a highlight → OK.

17) Alternate row shading (light gray)

Range: `A2:I{N}`
New Rule → Use a formula
Formula: `=ISEVEN(ROW())`
Format: Fill = light gray → OK.
(If you want shading to start from row 2 specifically regardless of sheet row numbers, you can also use `=MOD(ROW()-ROW($A$2),2)=0`*.)*

18) Admin dept hired after 1990

Range: `A2:I{N}`
New Rule → Use a formula
Formula: `=AND($E2="Admin",$H2>DATE(1990,1,1))`
Format: Choose a highlight → OK.

19) Salary above overall average (light blue)

Range: `A2:I{N}`
New Rule → Use a formula
Formula: `=$I2>AVERAGE($I:$I)`
Format: Fill = light blue → OK.
(If you prefer a fixed average for the data range only, use `AVERAGE($I$2:$I${N})`*.)*

20) Region = east and Salary < 9500 (red)

Range: `A2:I{N}`
New Rule → Use a formula
Formula: `=AND(LOWER($F2)="east",$I2<9500)`
Format: Fill = red → OK.

Advanced Excel Lab Assisgnment 9

Q.No	Questions
1	Find the Product Name for Product ID P005.
2	Retrieve the Product Price of Tablet (P010) using its Product ID.
3	Which Product ID corresponds to the price of ?18,000?
4	Using VLOOKUP, find the Product Price for Headphones (P015).
5	What will VLOOKUP return if you search for Product ID P999 (not in the list)?
6	Find the Product Name of the item priced at ?25,000.
7	Using Product ID P007, find the Product Price.
8	Get the Product ID of Keyboard using VLOOKUP.
9	Find the Product Name for Product ID P002.
10	Retrieve the Product Price for Product ID P014.
11	What will VLOOKUP return if the column index number is greater than the number of columns in the table?
12	Retrieve the Product Name for Product ID P008.

Data Assisgnment 9

S.No.	Product ID	Product Name	Product Price
1	P001	Laptop	55000
2	P002	Mouse	800
3	P003	Keyboard	1200
4	P004	Monitor	9000
5	P005	Printer	7000
6	P006	Scanner	6000
7	P007	Webcam	2500
8	P008	Headphones	1500
9	P009	Speaker	3500
10	P010	Microphone	2000
11	P011	External HDD	4500
12	P012	USB Drive	700
13	P013	Smartphone	30000
14	P014	Tablet	20000
15	P015	Smartwatch	12000
16	P016	Charger	1500
17	P017	Router	4000
18	P018	Projector	25000
19	P019	Graphics Card	18000
20	P020	RAM Module	6000

Expected Result

Q1. Find the Product Name for Product ID **P005**.
Answer: Printer
Formula:
`VLOOKUP("P005", B2:D16, 2, FALSE)`

Q2. Retrieve the Product Price of **Tablet (P010)**.
Answer: 15000
Formula:
`VLOOKUP("P010", B2:D16, 3, FALSE)`

Q3. Which Product ID corresponds to the price of **₹18,000**?
Answer: P012 (Smartwatch)
Formula:
`VLOOKUP(18000, D2:D16, 1, FALSE)`

Q4. Find the Product Price for **Headphones (P015)**.
Answer: 2000
Formula:
`VLOOKUP("P015", B2:D16, 3, FALSE)`

Q5. What will VLOOKUP return if you search for **P999**?
Answer: `#N/A`
Formula:
`VLOOKUP("P999", B2:D16, 2, FALSE)`

Q6. Find the Product Name of the item priced at **₹25,000**.
Answer: Smartphone (P013)
Formula:
`VLOOKUP(25000, D2:D16, 2, FALSE)`

Q7. Using Product ID **P007**, find the Product Price.
Answer: 6000
Formula:
`VLOOKUP("P007", B2:D16, 3, FALSE)`

Q8. Get the Product ID of **Keyboard**.
Answer: P003
Formula:
`VLOOKUP("Keyboard", C2:D16, 1, FALSE)`

Q9. Find the Product Name for Product ID **P002**.

Answer: Mouse
Formula:
```
VLOOKUP("P002", B2:D16, 2, FALSE)
```

Q10. Retrieve the Product Price for Product ID **P014**.
Answer: 12000
Formula:
```
VLOOKUP("P014", B2:D16, 3, FALSE)
```

Q11. What will VLOOKUP return if column index > number of columns?
Answer: `#REF!`
Example Formula:
```
VLOOKUP("P003", B2:D16, 5, FALSE)
```

Q12. Retrieve the Product Name for Product ID **P008**.
Answer: Scanner
Formula:
```
VLOOKUP("P008", B2:D16, 2, FALSE)
```

Advanced Excel Lab Assisgnment 10

Part A: Conceptual Questions

Q1 What is the difference between INDEX and VLOOKUP?

Q2 Explain how the MATCH function is used with INDEX.

Q3 Why is INDEX + MATCH preferred over VLOOKUP in many cases?

Part B: Practical Exercises (Use Employee_Data sheet)

Q1a Write an INDEX formula to return the Name of the 3rd employee in the list.

Q1b Use INDEX to return the Salary of the 5th employee.

Q2a Use MATCH to find the position of 'Finance' in the Department column.

Q2b Use MATCH to find the position of employee ID 104 in the Employee ID column.

Q3a Write a formula to find the Salary of 'Priya' using INDEX + MATCH.

Q3b Find the Department of 'Suresh' using INDEX + MATCH.

Q4 Create a search box where user types employee name, return Salary using INDEX + MATCH.

Part C: Challenge Question (Use Product_Data sheet)

Q1 Write a formula using INDEX + MATCH to find the Stock of 'Tablet'.

Q2 Create a dynamic lookup: if user types a Product ID, return its Price.

Employee_Data

Employee ID	Name	Department	Salary
101	Rahul	IT	50000
102	Priya	HR	45000
103	Ankit	Finance	55000
104	Neha	IT	52000
105	Suresh	Marketing	48000

Product_Data

Product ID	Product Name	Price	Stock
P001	Laptop	65000	12
P002	Mobile	25000	30
P003	Tablet	18000	20
P004	Headphones	2500	50
P005	Monitor	12000	15

Question	Answer Formula / Result
Q1a	INDEX(Name Range, 3) → Ankit
Q1b	INDEX(Salary Range, 5) → 48000
Q2a	MATCH('Finance', Department Range, 0) → 3
Q2b	MATCH(104, Employee ID Range, 0) → 4
Q3a	INDEX(Salary Range, MATCH('Priya', Name Range, 0)) → 45000
Q3b	INDEX(Department Range, MATCH('Suresh', Name Range, 0)) → Marketing
Q4	INDEX(Salary Range, MATCH(SearchName, Name Range, 0))
Challenge a	INDEX(Stock Range, MATCH('Tablet', Product Name Range, 0)) → 20
Challenge b	INDEX(Price Range, MATCH(InputID, Product ID Range, 0))